

Stock Analysis Model

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Stock Analysis Model — Overview

A quantitative framework that screens the entire US equity market every day and ranks companies on a disciplined value-investing framework (Graham / Buffett / Munger). For each company it produces a **BUY / LEAN BUY / HOLD / PASS** rating backed by a full valuation, quality, and moat breakdown.

How it works — the daily pipeline

1. **Universe.** Pulls the full list of US-listed equities (~9,000+ tickers) from SEC EDGAR.
2. **Phase 1 screen.** Two cheap filters run before any expensive analysis:
 - **Market cap $\geq$ \$300M** — drops micro-caps and shells that can't be meaningfully valued.
 - **ROIC $>$ WACC** — keeps only businesses earning a *positive economic spread* (they create value rather than destroy it).
3. **Phase 2 deep analysis.** Every survivor gets a full multi-model valuation, scoring, and rating.
4. **Sector enrichment.** Sector-specific KPIs are layered on so a bank isn't judged by the same yardstick as a software company.
5. **Outputs.** An interactive HTML report, an Excel workbook, and a JSON snapshot — the snapshot is archived every day so a full historical time series accumulates.

Latest run (2026-06-19): 2,193 companies fully analyzed from a starting universe of ~9,100 tickers.

Valuation engine

Each company is valued with several **independent** models, then cross-checked against one another for confidence:

Model	What it captures
DCF (discounted cash flow)	Owner-earnings intrinsic value, with Monte-Carlo simulation on key inputs
RIM (residual income model)	Book value plus the value of returns above the cost of capital
EPV (earnings power value)	Graham/Greenwald zero-growth “floor” value
DDM (dividend discount / H-model)	Fair value for established dividend payers
CAPM	Cost of equity and the WACC hurdle rate
Relative multiples	P/Fair-Value, P/FCF, P/Tangible-Book vs. peers
NAV (net asset value)	Asset-based value for REITs

Scoring framework

A **0–100 composite score** blends five weighted categories:

Category	Weight	What it measures
Moat	40%	ROIC spread vs. WACC, ROIC consistency & trend, gross & free-cash-flow margins
Valuation	20%	Margin of safety vs. DCF, P/FV, P/FCF, P/TBV, EPV floor, RIM margin of safety
Quality	20%	Interest coverage, net debt/EBITDA, accruals, cash conversion, ROE, Piotroski F-score
Growth	10%	Revenue & free-cash-flow durability, margin trend, fundamental growth
Ownership	10%	Shareholder yield, buyback rate, share-count direction (dilution vs. shrink)

Alongside the score, every company is graded against **26 pass/fail gates** spanning the same five categories — a fast “how many quality tests does it clear” read (e.g. *26/26*). A **Monte-Carlo uncertainty penalty** trims the composite when the fair-value estimate is wide, so speculative valuations can’t score as confidently as well-supported ones.

RATING BANDS

Rating	Composite score
BUY	≥ 60
LEAN BUY	≥ 43
HOLD	≥ 29
PASS	< 29

The model is deliberately **value-oriented**: it rewards high-quality businesses trading *below* intrinsic value, so its top picks tend to be out-of-favor laggards rather than momentum darlings. Daily validation confirms a *negative* correlation between composite score and trailing 12-month return — which is the expected, intended behavior for a value strategy.

Data sources

- **SEC EDGAR** — filings, XBRL financial statements, insider transactions, legal disclosures
- **yfinance** — prices and market data
- **FDIC BankFind** — bank call-report metrics
- **ClinicalTrials.gov** — drug/biotech pipeline depth
- **Finnhub / Tiingo / FMP** — supplementary fundamentals and news

Sector-aware KPIs

Generic ratios don't fairly compare across industries, so the model layers in metrics that actually distinguish leaders within each sector:

- **Banks** — net interest margin, efficiency ratio, CET1 capital, non-performing loans, deposit beta
- **REITs** — FFO growth, AFFO margin, debt-maturity wall, net asset value
- **Technology** — R&D intensity, stock-based comp as % of revenue, FCF margin ex-SBC, net cash, Rule of 40
- **Healthcare** — R&D intensity, FDA pipeline depth (active clinical trials)
- **Industrials** — capex intensity, backlog-to-revenue, book-to-bill
- **Consumer** — inventory days, cash-conversion cycle, brand/advertising spend

- **Energy / Utilities / Materials** — capex-to-D&A reinvestment discipline
 - **Insurers** — combined ratio, cost of float
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Outputs & cadence

- **Interactive HTML report** — sortable screening tables, per-company deep-dive panels, sector analysis, multi-stock price charts, and a personal flagging/watchlist feature.
 - **Excel workbook + daily JSON snapshot** — the snapshot is committed to a history branch each day, building a longitudinal dataset over time.
 - **Fully automated**, runs end-of-day.
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Roadmap

The accumulating daily snapshots are intended to train and calibrate the model over time, sharpening it toward actionable, real-money BUY / HOLD / PASS recommendations.

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